Sec 10.1 Vector Functions

We can define a vector function by putting parameterizations for x , y , z into a vector

That is if x=f(t) y= g(t) z=h(t) we can define a vector function as:

r(t)=<f(t) , g(t) , h(t)>

We call f(t) , g(t) , h(t) the component functions.

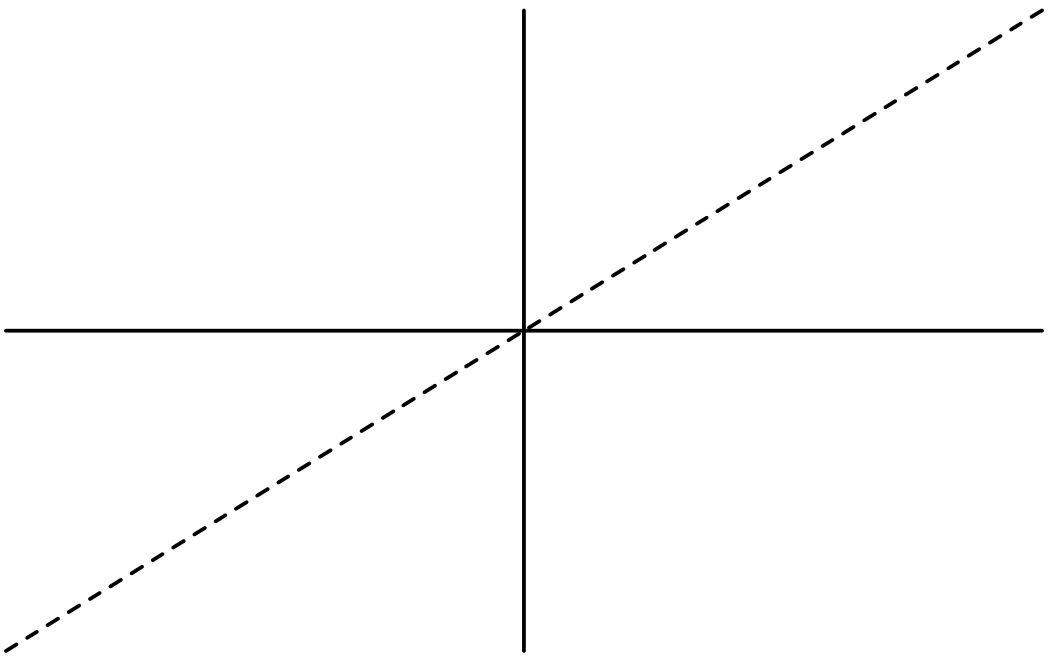
The domain of a vector function: This will be all of the t values we can plug in with no "mathematical issues" and which produce points that we can plot in 3d space.

Ex Find the domain of:

$\boldsymbol{r(t) = < \sqrt{t+2}, t^2+3, \frac{1}{t} >}$

Range of a vector function. For a vector function we get vectors as outputs.

Ex Sketch r(t)=<t,2t,3t>



The **limit** of a vector function $\mathbf{r}$ is defined by taking the limits of its component functions as follows.

1 If $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$, then

$$\lim_{t \to a} \mathbf{r}(t) = \left\langle \lim_{t \to a} f(t), \lim_{t \to a} g(t), \lim_{t \to a} h(t) \right\rangle$$

provided the limits of the component functions exist.

3–4 Find the limit.

3. $\lim_{t \to \infty} \left\langle \dfrac{1 + t^2}{1 - t^2}, \tan^{-1} t, \dfrac{1 - e^{-2t}}{t} \right\rangle$

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Finding the vector equation of a line.

Hw example

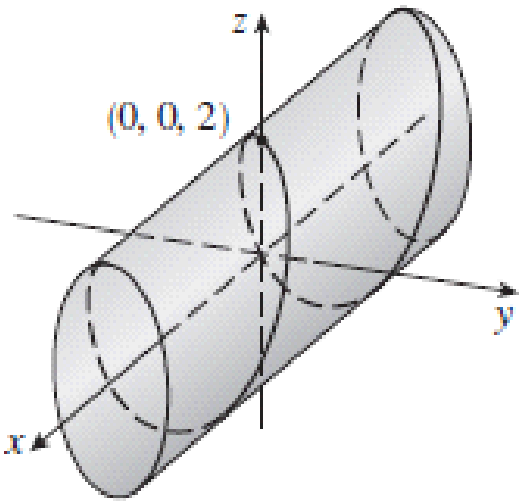
15–18 Find a vector equation and parametric equations for the line segment that joins P to Q.

15. $P(0, 0, 0)$, $Q(1, 2, 3)$

Using projections on to the coordinate planes to sketch a curve.

13–14 Draw the projections of the curve on the three coordinate planes. Use these projections to help sketch the curve.

13. $\mathbf{r}(t) = \langle t, \sin t, 2\cos t \rangle$



25. Show that the curve with parametric equations $x = t \cos t$, $y = t \sin t$, $z = t$ lies on the cone $z^2 = x^2 + y^2$, and use this fact to help sketch the curve.